skaters: Model First, Conform Last — A Composition-Based Automatic Online Distributional Forecaster

Peter Cotton
Micropediction
peter.cotton@microprediction.com
https://skaters.microprediction.org

June 19, 2026

Abstract. The Python package **skaters** performs online univariate time-series forecasting in which every prediction is a full probability distribution rather than a point. A forecaster is built by composition: invertible transforms chain together above a single distributional leaf, and ensembles combine such chains. The leaf fits its shape by optimising a proper scoring rule, so its objective is a choice rather than a fixed property of the method. This separates two concerns the forecasting and conformal-prediction literatures tend to merge: fitting the model, judged by likelihood, and conforming the predictive tail to a downstream score such as CRPS. We call the arrangement *model first, conform last*. The library is written twice, in pure Python and in zero-dependency JavaScript that agrees to within 10^{-6}, so a model runs unchanged on a server or in a browser. We evaluate it against classical, neural, and pretrained foundation-model baselines on FRED series.

Keywords: probabilistic forecasting, time series, proper scoring rules, conformal prediction, online learning, Python, JavaScript, **skaters**.

1 Introduction

Most deployed forecasters return a point and, perhaps, an interval bolted on afterwards. Yet the quantity a decision-maker actually needs is the *predictive distribution*: the mean for an estimate, the spread for risk, the tails for position sizing, and a density for likelihood-based evaluation and combination. Two strands of recent work take distributions seriously but each gives something up.

Classical state-space and exponential-smoothing models [Hyndman et al., 2002, Hyndman and Khandakar, 2008] are distributional but rarely *online and composable*: adding a transform usually means re-deriving the filter. Conditional-heteroscedasticity models [Engle, 1982, Bollerslev, 1986] capture the heavy tails and volatility clustering of financial data, but commit to a single parametric form and are refit in batch. At the other extreme, conformal prediction [Vovk et al., 2005, 2019] is distribution-free and online, but a conformal predictive system outputs a cumulative distribution function, not a density. It is calibrated for coverage and CRPS but cannot be scored on log-likelihood at all, and an empirical conformal CDF assigns zero density — and hence $-\infty$ log-likelihood — to any value outside the observed residual range. Modern neural and foundation models [Salinas et al., 2020, Ansari et al., 2024, Das et al., 2024, Woo et al., 2024] are powerful distributional learners, but are built for global cross-series training on hardware accelerators, not for a single univariate stream running in a browser.

skaters takes a third route. Every node — transform, ensemble, or leaf — is a function `(y, state) -> ([Dist], state)` that consumes one observation and emits a list of `Dist` objects, one per forecast horizon, where a `Dist` is a weighted Gaussian mixture. Because the type is closed under composition, an entire model is a chain of invertible transforms with one distributional leaf at the bottom, and ensembles are simply skaters that combine other skaters. This single abstraction yields three contributions developed in turn:

1. *A pluggable objective* (Sections 3–4). The leaf fits its mixture weights by a proper scoring rule. Pointed at log-likelihood it models the conditional distribution directly; with its terminal copy pointed at CRPS while the trunk stays likelihood-weighted, the result is *model first, conform last*, which is competitive with a CRPS specialist on CRPS and stronger on likelihood.

2. *Distributional structure that survives composition* (Sections 5–7): a mean-preserving lattice projection that places atoms on the exact values a series revisits without moving the ensemble mean and vanishes on continuous data; an online coordinate search over whether a series is simple in a log, root, or linear coordinate; and a committed martingale model that learns only a volatility clock.

3. *A benchmark against eight distributional baselines* (Section 8), chosen as the strongest available opponents: the heavy-tail state of the art (**GARCH-t**), a neural density model (**NeuralForecast**), and **AutoARIMA** paired with split- and adaptive-conformal residuals — the CRPS-optimised methods that are hardest to beat on that metric. Every method is scored on held-out log-likelihood and CRPS through a single, identical `Dist`-based protocol.

The unifying premise is the No-Free-Lunch theorem [Wolpert and Macready, 1997] taken seriously: every "method" is a prior, and averaged over all series none dominates. The right response is not to pick one but to make the priors explicit, weight them online, and bound the cost of a wrong prior by how fast the weighting abandons it.

2 The `Dist` abstraction and composable transforms

A `Dist` is a weighted mixture $\sum_i w_i \mathcal{N}(\mu_i, \sigma_i^2)$ with $\sigma_i > 0$ and $\sum_i w_i = 1$. It exposes `mean`, `std`, `pdf`, `cdf`, `logpdf`, `quantile`, and a closed-form `crps`, together with the affine maps (`shift`, `scale`, `affine`) and a moment-matching `prune` needed to propagate it through transform inverses and to bound component growth in ensembles. The Gaussian mixture is a deliberate choice: it is closed under affine maps and mixing, admits a closed-form CRPS through the elementary identity $\mathbb{E}\,|\mathcal{N}(m, s^2)| = 2s\,\phi(m/s) + m\,(2\Phi(m/s) - 1)$, and approximates heavy tails arbitrarily well as a scale mixture.

A *transform* is an online bijection given by a `forward` (scalar in, scalar out) and an `inverse_k` that maps k `Dist`s in the transformed space back to the original. Differencing, exponential smoothing, drift, Holt's linear trend, fractional differencing, autoregression (via online recursive least squares), **GARCH**-style scaling, seasonal differencing, power and Yeo–Johnson coordinates [Box and Cox, 1964, Yeo and Johnson, 2000], and standardisation are all transforms. *Conjugation* composes a transform with an inner skater,

$$y \xrightarrow{T} y' \to \text{inner} \to D' \xrightarrow{T^{-1}} D,$$

and because every step is online and the inverse propagates the full mixture, multi-step uncertainty accumulates correctly — for example variance grows as $\sum_h \sigma_h^2$ under differencing.

Numerical note. Mixture variance is computed in centred form, $\mathrm{Var} = \sum_i w_i\,(\sigma_i^2 + (\mu_i - \bar{\mu})^2)$, rather than $\mathbb{E}[X^2] - \mathbb{E}[X]^2$. The latter catastrophically cancels when a tight component sits at a large mean, silently producing $\sigma = 0$ and an invalid predictive. Running moments use Welford's recurrence [Welford, 1962].

3 The terminal-leaf ensemble: model first

Bayesian model averaging over a heterogeneous pool [Hoeting et al., 1999, Raftery et al., 2005] combines full predictive distributions, which preserves the combined mean and variance but *washes out higher moments*: a heavy-tailed leaf used inside the pool collapses toward a Gaussian shape at the output, because a mixture of differently located Gaussians is dominated by the spread of their means. We therefore use the candidate models only as mean forecasters, weight them online by predictive likelihood (with a learning-rate shrinkage and a per-depth complexity penalty, in the spirit of gradient boosting), combine their means, and model the distribution of the combined residual with a single terminal leaf:

$$y \;\to\; \hat{\mu} = \sum_i w_i\,\hat{\mu}_i, \qquad r = y - \hat{\mu} \;\to\; \text{leaf} \;\to\; D, \qquad \hat{D} = D \text{ shifted by } \hat{\mu}.$$

Because exactly one leaf reaches the output, its shape — heavy tails, learned online — survives undiluted. This is the *model first* half of the recipe: a likelihood-weighted trunk whose only job is to get the conditional mean right. The candidate pool spans exponential smoothing, differencing, drift, Holt's linear trend, autoregression, grouped long-lag autoregression, fractional differencing, a seasonal block, a Yeo–Johnson coordinate grid (Section 6), and a fast/slow two-systems block.

4 Metric-agnostic leaves: conform last

The leaf is itself a fixed Gaussian *scale mixture* — components $\mathcal{N}(0, c_k\,\hat{\sigma})$ over a fixed log-spaced dictionary $\{c_k\}$, with $\hat{\sigma}$ an exponentially weighted scale — whose only learned quantities are the mixture weights $\{w_k\}$ on the simplex. A Gaussian scale mixture is a Student-t in the limit, so the leaf approximates heavy tails by construction while remaining a plain `Dist`. The weights are fit online by a proper scoring rule [Gneiting and Raftery, 2007]:

- `likelihood` — recency-weighted expectation–maximisation on the component responsibilities;
- `crps` — exponentiated-gradient descent on the simplex minimising the closed-form mixture CRPS [Matheson and Winkler, 1976]; the gradient is exact, since the mixture CRPS is a finite sum of $\mathbb{E}\,|\mathcal{N}(m, s^2)|$ terms.

Placing the CRPS leaf at the terminal of a likelihood-weighted trunk gives *model first, conform last.* The trunk's job is to model, and the right objective for modelling is likelihood, a proper, tail-sensitive score that is, up to a constant, the Kelly log-growth criterion [Kelly, 1956]; corrupting it to chase CRPS would repeat the conformal mistake of discarding the density. The tail's job is to be shaped, and that is the only place a downstream metric (CRPS) should get a vote. Section 9 shows this configuration improving on a pure-likelihood policy on CRPS and on likelihood alike: a CRPS-fit leaf is more outlier-robust, so it even generalises slightly better in log-likelihood.

5 The lattice projection

Administrative and grid-quoted series — policy rates, posted prices, pegged exchange rates — revisit a small set of exact values, and a continuous predictive density cannot place mass there. We add a *projection*: a recency-weighted frequency table of exact values, with near-Dirac atoms on the values revisited above a noise floor, each carrying its observed frequency as probability. The atom is mean-preserving: the continuous part is recentred so the atoms add mass without moving the ensemble mean.

Two properties make this a safe default. First, on continuous data nothing is revisited, no atom fires, and the projection vanishes identically — it is free when unused. Second, unlike a last-value spike it captures values that recur often but never consecutively (a rate that repeatedly returns to a round number). The pull of the level (its tendency to persist) is left to the trunk, where the random-walk candidate earns its weight by likelihood, so the projection remains a pure statement about mass, not mean.

6 Coordinate learning

Whether a series is simple in a linear, log/multiplicative, or root coordinate is itself learnable. We add a coarse grid of Yeo–Johnson λ candidates — the signed Box–Cox family [Box and Cox, 1964, Yeo and Johnson, 2000], defined on all of $\mathbb{R}$ — to the pool, so the ensemble learns the coordinate online rather than committing up front. This is No-Free-Lunch-safe: a wrong coordinate is simply down-weighted by likelihood. The same mechanism expresses a non-negativity prior at $\lambda = 0$ (the log coordinate), though we note that because the `Dist` is a Gaussian mixture the inverse uses the delta-method linearisation, so non-negativity is approximate rather than exact.

7 A martingale with a learned volatility clock

For near-martingale *levels* we provide a committed model, `doob`: the mean is pinned to the last value (a driftless martingale) and only the volatility clock is learned, as a Bayesian average over martingale predictives that differ in their volatility model (constant, **GARCH**, slowly varying, heavy-tailed). Because every candidate shares the same mean, plain averaging blends the clocks *without* washing out kurtosis — the committed mean is exactly what makes model averaging the right tool here, in contrast to Section 3. By the Dambis–Dubins–Schwarz theorem [Dambis, 1965, Dubins and Schwarz, 1965, Doob, 1953] a continuous martingale is a time-changed Brownian motion, so the model is literally "Brownian motion on a stochastic clock". It beats the general forecaster on near-martingale levels by committing the mean and spending its capacity on the clock; on genuinely mean-reverting series the martingale prior is wrong and it gives ground — a sharp, single-purpose instrument, fed *levels* rather than pre-differenced changes.

8 Experiments

Universe. To avoid a hand-picked series list we draw from the Federal Reserve Economic Data [Federal Reserve Bank of St. Louis], taking cached daily series and auto-transforming each to its one-step *changes* (log-difference for strictly positive levels, else first difference) — the stationary-ish innovation stream where heavy tails and regime shifts live, and which keeps log-likelihood comparable across series of different scale. We keep the last 1000 changes per series and score the last 300 as a rolling one-step-ahead test window after a burn-in.

Baseline	LL (all / cont.)	CRPS (all / cont.)	Mean LL (cont.)
AutoARIMA	**78 / 64 %**	51 / 47 %	2.09
AutoETS	**92 / 88 %**	68 / 67 %	2.20
SARIMAX (statsmodels)	**82 / 72 %**	53 / 49 %	2.13
ETS (statsmodels)	**93 / 89 %**	69 / 68 %	2.12
GARCH$(1,1)$-t	**79 / 67 %**	76 / 66 %	**2.72**
NF-Student$t$	**100 / 100 %**	78 / 76 %	0.82
AutoARIMA + conformal	**84 / 75 %**	37 / 29 %	1.47
AutoARIMA + ACI	**87 / 80 %**	36 / 28 %	1.89

Table 1: Per-series win-rate of `laplace` versus each baseline on 500 FRED change-series, all and on the 309 continuous series. LL counts series where `laplace` has the higher held-out log-likelihood; CRPS where it has the lower CRPS. The final column is each baseline's mean continuous log-likelihood (`laplace`: 2.855). Higher is better for LL; lower for CRPS.

8.1 Against eight distributional baselines

On 500 such series we compare the general forecaster `laplace` against every distributional one-step baseline we could assemble: **statsforecast**'s [Garza et al., 2022] **AutoARIMA** and **AutoETS**; **statsmodels SARIMAX**$(1,0,1)$ and simple exponential smoothing, which emit *exact* closed-form Gaussian predictives; a constant-mean **GARCH**$(1,1)$ model with Student-t innovations [Bollerslev, 1986, 1987] from the **arch** package, the classical heavy-tail / volatility-clustering state of the art; a **NeuralForecast** [Olivares et al., 2022] multilayer perceptron with a Student-t distribution head; and **AutoARIMA** paired with conformal residual quantiles [Vovk et al., 2019] in both a split-conformal and an adaptive-conformal (ACI, Gibbs and Candès, 2021) variant.

The protocol is a *fair rolling one-step-ahead*: each baseline is fit on an expanding or rolling window with periodic refit, and every method — ours and theirs — is turned into the *same* `Dist` and scored on held-out log-likelihood *and* CRPS over the same test window. The parametric Student-t predictives (**GARCH**, **NeuralForecast**) are represented as finite Gaussian scale mixtures, matching the analytic t density to $\sim 10^{-3}$. We report per-series win-rates (the fraction of series on which `laplace` scores better), and additionally restrict to the 309 continuous series — those with under 5% exactly-repeating one-step changes — since on repeating/grid series the lattice projection (Section 5) gives a large but metric-specific advantage that would otherwise dominate the aggregate. Table 1 reports the result.

The likelihood race. `laplace` wins the per-series log-likelihood race against all eight baselines, on the full universe and on the continuous subset, including the two exact-Gaussian references (**SARIMAX**, statsmodels ETS) for which there is no band-reconstruction approximation to blame.

GARCH-t is the tightest case. The **GARCH**$(1,1)$-t model is purpose-built for the heavy tails and volatility clustering of financial change-series, so it is the strongest opponent on log-likelihood, and the numbers bear that out. It is the closest race (67% of continuous series), and its mean continuous log-likelihood, 2.72, is nearest to ours, 2.855, a narrow edge of about 0.14 nats per observation. That the advantage holds against the heavy-tail specialist is the substantive point: the general forecaster captures the same conditional heteroscedasticity without committing to a parametric volatility law.

Model	density	LL (all / cont.)	CRPS (all / cont.)	Mean LL (cont.)
Moirai (1.1-R-small)	native t	98 / **97** %	94 / 93 %	0.92
Lag-Llama	native t	67 / **99** %	65 / 94 %	0.39
Chronos-Bolt (small)	quantile*	76 / **100** %	67 / 88 %	−1.96
TimesFM (2.5-200M)	quantile*	75 / **100** %	58 / 72 %	−1.18

Table 2: Per-series win-rate of `laplace` versus four zero-shot foundation models on 120 change-series (69 continuous). *Chronos-Bolt/TimesFM log-likelihood is reconstructed from a quantile head and is tail-limited; read their CRPS as the fairer signal. `laplace` mean continuous logpdf: 1.51.

The neural result, in context. `laplace` beats NF-Student*t* on 100% of series, but the comparison needs reading carefully. NF here is **NeuralForecast** run in the matched online univariate one-step protocol, a small per-series multilayer perceptron with periodic refit, which is its weak regime. Neural and foundation forecasters [Salinas et al., 2020, Olivares et al., 2022] are built for global cross-series training, where they amortise structure across thousands of related series. The result says nothing about **DeepAR**-style models in that setting, and we do not claim it does; it says only that a single short univariate stream does not give a neural density head enough to beat an online ensemble. The NF densities are well formed (median log-likelihood 0.50, none floored), so the 100% is a real calibration gap, not a degenerate score.

CRPS. As ever, CRPS tells a mixed story. `laplace` beats the smoothing baselines (**AutoETS**, ETS), **GARCH**-*t* and NF on CRPS, roughly ties the **ARIMA** family (**AutoARIMA** 51%, **SARIMAX** 53%), and *loses* to the conformal variants (37%, 36%), which are CRPS-optimised by construction. This is exactly the thesis of Sections 4: likelihood is the metric where a faithful density wins; CRPS is conformal's home turf, where we are competitive but not dominant. A practitioner who truly only cares about CRPS can repoint the leaf (Section 9); a practitioner who cares about the density — for risk, sizing, or combination — should prefer the log-likelihood column.

Scope. Five hundred change-series, one-step horizon. Prophet, multi-step horizons and raw levels are left for later; zero-shot foundation models get their own protocol in Section 8.2.

8.2 Against zero-shot foundation models

Pretrained time-series foundation models use a fundamentally different protocol: they are applied *zero-shot*, conditioning on a context window with no fitting. We evaluate four — Chronos-Bolt [Ansari et al., 2024], TimesFM 2.5 [Das et al., 2024], Moirai [Woo et al., 2024], and Lag-Llama [Rasul et al., 2023] — on 120 change-series (69 continuous). At each step the model receives a fixed 256-length window of the preceding changes and predicts the next change, with all windows for a series batched into one inference call. Each predictive is turned into the same `Dist` — the native Student-t head for Moirai/Lag-Llama, a sample/quantile reconstruction for Chronos-Bolt/TimesFM — and `laplace` is re-scored on the identical window. Table 2 reports per-series win-rates.

On the continuous series `laplace` beats all four foundation models zero-shot on both metrics (97–100% LL, 72–94% CRPS). The native-density models (Moirai, Lag-Llama) are the meaningful likelihood opponents and the closest, but still lose per-series. On repeat-heavy/grid series the foundation models are competitive or better — Lag-Llama even beats `laplace` on mean log-likelihood over the *full* universe (6.54 vs 4.52) by placing tight mass on revisited values, the same effect as the lattice projection — which is why the continuous split matters. The scope is narrow: zero-shot, no refit, fixed context, and forecasting the change stream rather than the levels these models were chiefly trained on.[1]

9 Ablations: repointing the objective

A companion sweep on 2,501 FRED series isolates the effect of the terminal-leaf objective against the **crepes** conformal predictive system (given its three best calibration windows and scored on its own CDF via the pinball decomposition of CRPS). Because the conformal opponent there uses only a naive mean, we read this as an ablation of our own objective knob rather than a state-of-the-art claim. Reporting per-policy and de-correlating to one vote per correlated family (195 families):

Forecaster (identical structure)	CRPS, raw	CRPS, family	Mean LL
likelihood trunk + likelihood leaf	82.6 %	48.7 %	2.96
likelihood trunk + CRPS leaf	89.5 %	**60.1 %**	**3.01**
bare CRPS leaf (no trunk)	80.8 %	60.3 %	2.96

Table 3: Effect of the terminal-leaf objective. The middle row is the *model first, conform last* recipe. A general likelihood policy is essentially even with conformal on CRPS over independent families (48.7%); repointing only the terminal leaf to CRPS lifts that to 60.1% and raises log-likelihood, matching the dedicated CRPS specialist while modelling faithfully.

Three further observations motivate the small public API. Priors wash out: across 25 series with ≥ 2000 changes the conventional prior-flavoured policies (trend, long-memory, drift, minimax, fast/slow) sit within 0.0006 nats of the uniform-prior general forecaster on the second half of each series, and the first-half winner beats the general forecaster on the second half only 44% of the time, worse than chance. Conform-last is near-free: on idealised light-tailed processes the CRPS leaf costs 0.005–0.007 nats versus the likelihood leaf, while on heavy-tailed and stochastic-volatility processes it helps (up to +0.04); real data lives in the second regime. And the lattice projection is free when unused. Together these justify exposing exactly one general forecaster (uniform prior, CRPS leaf, lattice projection and coordinate learning on by default) plus one committed martingale specialist; the remaining structural ideas do not pay their way out of sample.

10 Discussion

The unifying theme is No-Free-Lunch [Wolpert and Macready, 1997] taken seriously. Every "method" is a prior; averaged over all series none dominates. The correct response is not to choose one but to make the priors explicit, weight them online, and bound the cost of a wrong prior by how fast the weighting abandons it. Read that way, the named

[1] Per-series *fine-tuning* does not rescue the foundation models on this task: naively fine-tuning Lag-Llama on a single series' history (5 epochs) *catastrophically overfits*, collapsing its held-out log-likelihood from +1.5 to below −100 while taking ∼15× longer. This is the expected failure mode — adapting a pretrained model to one short univariate stream is not its design regime; zero-shot (or domain-level fine-tuning across many series) is the intended use. We therefore report the zero-shot result as the headline.

methods of classical forecasting are convenience bundles, and the natural object is a single adaptive ensemble with two orthogonal terminal knobs — *which proper scoring rule the leaf optimises*, and *whether a lattice projection is active* — neither of which is a prior over the trunk. The empirical headline supports the design: against the strongest distributional opponents available on CPU, including the heavy-tail and neural state of the art, the general forecaster wins the likelihood race, and where it does not lead — CRPS — the deficit is confined to methods explicitly optimised for that metric and is recoverable by repointing a single knob.

11 Heritage and reproducibility

skaters is a from-scratch rewrite distilling ideas from **timemachines** [Cotton, 2021], an earlier competition-tested online-forecasting package, and builds on years of running live distributional-prediction contests at Microprediction, where forecasts are scored continuously as full distributions. The library is implemented twice — pure Python and zero-dependency JavaScript — and a parity suite checks roughly 90,000 probe values to 10^{-6} on every run, so results are reproducible across both runtimes and in the browser via **Pyodide** [The Pyodide development team, 2021]. The benchmark of Section 8 is a single parallel script (`benchmarks/sota_study.py`) that scores every method through the identical `Dist` interface; given a FRED key and the optional baselines (**statsforecast**, **arch**, **statsmodels**, **neuralforecast**) it reproduces Table 1 end to end.

References

Abdul Fatir Ansari et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024.

Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

Tim Bollerslev. A conditionally heteroskedastic time series model for speculative prices and rates of return. *The Review of Economics and Statistics*, 69(3):542–547, 1987.

George EP Box and David R Cox. An analysis of transformations. *Journal of the Royal Statistical Society: Series B*, 26(2):211–243, 1964.

Peter Cotton. timemachines: Evaluation of online time-series forecasting algorithms. `https://github.com/microprediction/timemachines`, 2021.

KE Dambis. On the decomposition of continuous submartingales. *Theory of Probability & Its Applications*, 10(3):401–410, 1965.

Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. *International Conference on Machine Learning*, 2024.

Joseph L Doob. *Stochastic Processes*. Wiley, 1953.

Lester E Dubins and Gideon Schwarz. On continuous martingales. *Proceedings of the National Academy of Sciences*, 53(5):913–916, 1965.

Robert F Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

Federal Reserve Bank of St. Louis. Federal reserve economic data (FRED). https://fred.stlouisfed.org.

Federico Garza, Max Mergenthaler Canseco, Cristian Challu, and Kin G Olivares. StatsForecast: Lightning fast forecasting with statistical and econometric models. *PyCon Salt Lake City, Utah, US 2022*, 2022. Nixtla.

Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, 34:1660–1672, 2021.

Tilmann Gneiting and Adrian E Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

Jennifer A Hoeting, David Madigan, Adrian E Raftery, and Chris T Volinsky. Bayesian model averaging: a tutorial. *Statistical Science*, 14(4):382–401, 1999.

Rob J Hyndman and Yeasmin Khandakar. Automatic time series forecasting: the forecast package for R. *Journal of Statistical Software*, 27(3):1–22, 2008.

Rob J Hyndman, Anne B Koehler, Ralph D Snyder, and Simone Grose. A state space framework for automatic forecasting using exponential smoothing methods. *International Journal of Forecasting*, 18(3):439–454, 2002.

John L Kelly. A new interpretation of information rate. *The Bell System Technical Journal*, 35(4):917–926, 1956.

James E Matheson and Robert L Winkler. Scoring rules for continuous probability distributions. *Management Science*, 22(10):1087–1096, 1976.

Kin G Olivares, Cristian Challu, Federico Garza, Max Mergenthaler Canseco, and Artur Dubrawski. NeuralForecast: User friendly state-of-the-art neural forecasting models. *PyCon Salt Lake City, Utah, US 2022*, 2022. Nixtla.

Adrian E Raftery, Tilmann Gneiting, Fadoua Balabdaoui, and Michael Polakowski. Using Bayesian model averaging to calibrate forecast ensembles. *Monthly Weather Review*, 133 (5):1155–1174, 2005.

Kashif Rasul et al. Lag-Llama: Towards foundation models for probabilistic time series forecasting. *arXiv preprint arXiv:2310.08278*, 2023.

David Salinas, Valentin Flunkert, Jan Gasthaus, and Tim Januschowski. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3):1181–1191, 2020.

The Pyodide development team. Pyodide: Python with the scientific stack, compiled to WebAssembly. https://pyodide.org, 2021.

Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

Vladimir Vovk, Ilia Nouretdinov, Valery Manokhin, and Alexander Gammerman. Conformal predictive distributions with kernels. In *Conformal and Probabilistic Prediction and Applications*, 2019.

BP Welford. Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3):419–420, 1962.

David H Wolpert and William G Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.

Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. *International Conference on Machine Learning*, 2024.

In-Kwon Yeo and Richard A Johnson. A new family of power transformations to improve normality or symmetry. *Biometrika*, 87(4):954–959, 2000.